

Python Operators and Expressions:

Arithmetic operators

```
In [1]: a=5  
b=3      # sum= + , diff= -, product= *, div= /  
  
a+b , a-b , a*b , a/b
```

```
Out[1]: (8, 2, 15, 1.6666666666666667)
```

Comparison operators

```
In [2]: a == b      # equal operator a=5, b= 3
```

```
Out[2]: False
```

```
In [3]: a != b      # not equal to
```

```
Out[3]: True
```

Logical operators

```
In [4]: a>0 and b>0 # logical and(&)
```

```
Out[4]: True
```

```
In [5]: a> 0 | b<0  # logical or (/)
```

```
Out[5]: False
```

```
In [6]: not a>0
```

```
Out[6]: False
```

Unary Operators: Operate on a single operand. (a)

```
In [9]: -10,+11,-12,+13 # symbol (-)unary , (+)unary
```

```
Out[9]: (-10, 11, -12, 13)
```

```
In [19]: r=[1,2,3]  
r1=[1,2,3]  
print(r is r1)  
print(r == r1)
```

```
False
```

```
True
```

```
In [24]: n1=5      # bit and & operator # 0101 & 0011 = 0001 (1 in decimal)  
n2=3  
n1&n2
```

```
Out[24]: 1
```

```
In [25]: n1|n2 # bit or (/) # 0101 | 0011 = 0111 (7 in decimal)
```

```
Out[25]: 7
```

```
In [26]: n1^n2 # Example with bitwise XOR (^) operator # 0101 ^ 0011 = 0110
```

```
Out[26]: 6
```

the membership operators in and not in are used to test whether a value is present in a sequence (such as a list, tuple, or string) or not.

```
In [37]: fruits = ['apple', 'banana', 'orange']
         'banana' in fruits, 'grapes' not in fruits
```

```
Out[37]: (True, True)
```

```
In [38]: x = [1, 2, 3]
         y = [1, 2, 3]
         z=x
         x is y, x is z, x is not y
```

```
Out[38]: (False, True, True)
```

e ternary conditional operator in Python.

```
In [71]: input=0
         int=int()
```

```
In [72]: hott_str= input("temp:")
         hottness=int(hott_str)
         mood = 'tempt' if hottness > 99 else 'not tempt'

         print(mood)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[72], line 1
----> 1 hott_str= input( )
      2 hottness=int(hott_str)
      3 mood = 'tempt' if hottness > 99 else 'not tempt'

TypeError: 'int' object is not callable
```

```
In [63]: # With a helpful prompt
         name = input("What is your name? ")
         print("Hello,", name)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[63], line 2
      1 # With a helpful prompt
----> 2 name = input( )
      3 print("Hello,", name)

TypeError: 'str' object is not callable
```

```
In [74]: # To keep only specific variables  
%reset_selective -f [variable_names_to_keep]
```

```
In [75]: hott_str= input("temp:")  
hottness=int(hott_str)  
mood = 'tempt' if hottness > 99 else 'not tempt'  
  
print(mood)
```

tempt

```
In [ ]:
```